

Complete Notes on Time Series Forecasting in Machine Learning

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September 07, 2025

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1 Introduction to Time Series Forecasting

A time series is a sequence of data points collected at regular intervals over time, such as daily stock prices, monthly sales, or hourly temperature readings. The goal of time series forecasting is to analyze historical data to predict future values, which is critical for planning and decision-making in various fields.

- **Business:** Forecasting sales to manage inventory or predict revenue.
- **Finance:** Predicting stock prices or market trends.
- **Weather:** Estimating future temperatures or rainfall.
- **Example:** A retailer uses past sales data to predict next quarters demand, ensuring optimal stock levels.

Visualization: A line plot with time on the x-axis (e.g., days 1 to 30) and values (e.g., sales) on the y-axis shows trends and patterns. The plot below demonstrates a time series with a slight upward trend and periodic fluctuations.

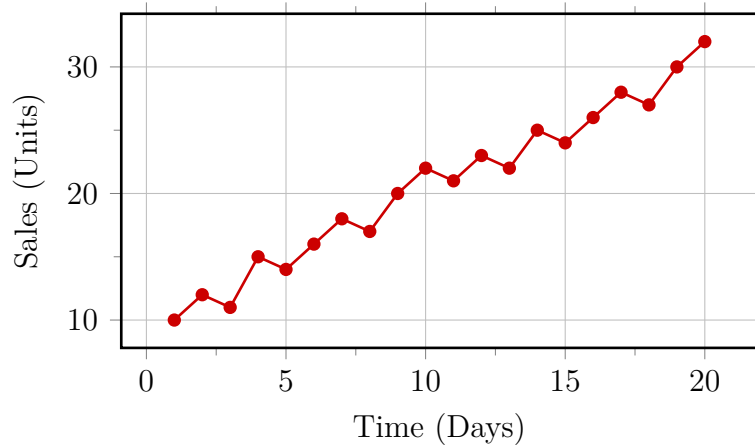


Figure 1: Example Time Series: Daily Sales

2 Components of Time Series

Time series data typically consists of four components that create the observed patterns:

- **Trend:** The long-term direction of the data, such as a steady increase in sales over years.
- **Seasonality:** Repeating patterns at fixed intervals, like higher sales every December.
- **Cyclical:** Fluctuations over longer periods without fixed intervals, such as economic booms and busts.
- **Irregular (Noise):** Random variations that cannot be modeled, like a sudden sales drop due to an unexpected event.

Additive Decomposition Formula:

$$Y_t = T_t + S_t + C_t + I_t$$

where:

- Y_t : Observed value at time t .
- T_t : Trend component.
- S_t : Seasonal component.
- C_t : Cyclical component.
- I_t : Irregular (noise) component.

Example Calculation: Suppose we have monthly sales data: [100, 120, 140, 160, 180]. Assume a linear trend $T_t = 90 + 10t$, seasonal component repeating every 4 months [10, 5, -5, -10], and no cyclical component. Calculate components for $t = 3$:

- Trend: $T_3 = 90 + 10 \cdot 3 = 120$.
- Seasonal: $S_3 = -5$ (third position in [10, 5, -5, -10]).
- Cyclical: $C_3 = 0$.
- Irregular: $I_3 = Y_3 - (T_3 + S_3) = 140 - (120 + (-5)) = 140 - 115 = 25$.

Thus, $Y_3 = 120 + (-5) + 0 + 25 = 140$.

3 Preprocessing Time Series Data

Preprocessing ensures data is clean and suitable for modeling. Key steps include handling missing values, outliers, and ensuring stationarity.

3.1 Handling Missing Values

Fill missing data to avoid model errors:

- **Mean Imputation:** Replace with the average of the series.
- **Forward Fill:** Use the previous value.
- **Example:** Data = [10, NaN, 14, 16]. Mean = $(10+14+16)/3 = 13.33$. Imputed: [10, 13.33, 14, 16].

3.2 Detecting and Handling Outliers

Outliers are extreme values that can skew models.

- **Method:** Identify values beyond 3 standard deviations from the mean.
- **Example:** Data = [10, 12, 14, 100, 18]. Mean = 30.8, Std = 37.3. Threshold = $30.8 \pm 3 \cdot 37.3 = [-81.1, 142.7]$. 100 is within bounds, but if it were 200, it would be outside.

3.3 Ensuring Stationarity

Stationary data has constant mean and variance. Non-stationary data (with trends or seasonality) needs transformation.

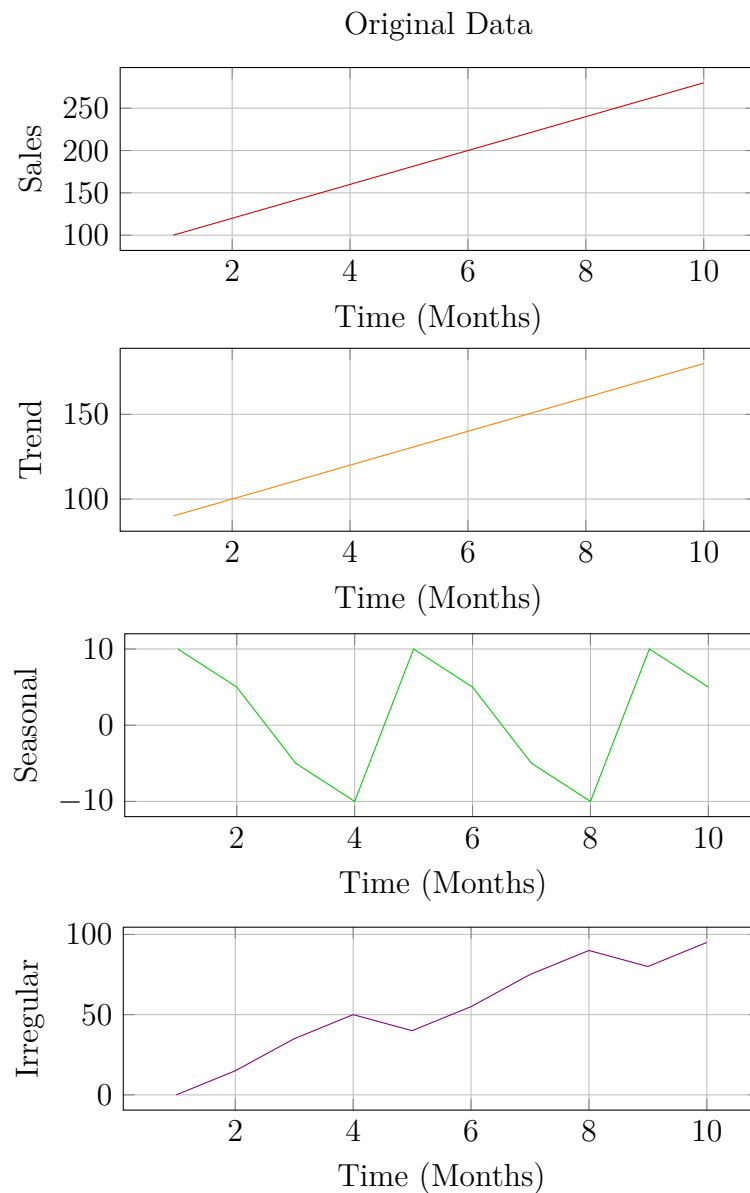


Figure 2: Time Series Decomposition

- **Test:** Augmented Dickey-Fuller (ADF). If p-value > 0.05 , data is non-stationary.
- **Differencing:** Subtract previous value to remove trend.
- **Formula:** $\Delta Y_t = Y_t - Y_{t-1}$.
- **Example Calculation:** Data = [10, 12, 15, 19, 24]. Difference = [NaN, 12-10=2, 15-12=3, 19-15=4, 24-19=5] = [NaN, 2, 3, 4, 5].

Moving Average Smoothing:

$$MA_t = \frac{1}{k} \sum_{i=0}^{k-1} Y_{t-i}$$

Example (k=3): Data = [10, 12, 14, 16, 18]. MA = [NaN, NaN, (10+12+14)/3=12, (12+14+16)/3=14, (14+16+18)/3=16].

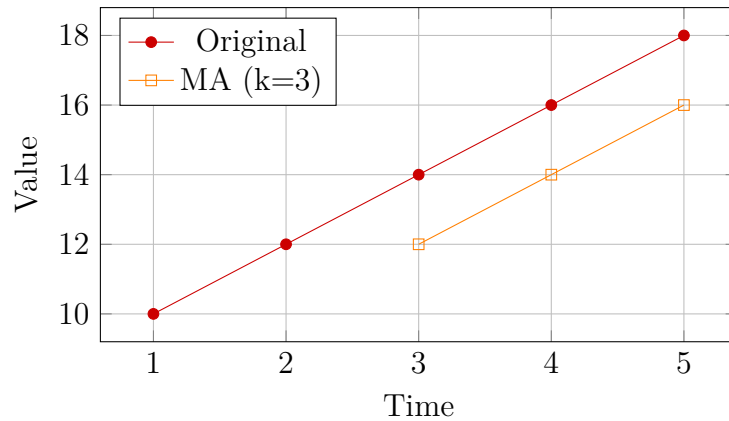


Figure 3: Moving Average Smoothing

4 Classical Forecasting Methods

Classical methods are simple and effective for linear patterns.

4.1 Moving Average

Predicts the next value as the average of the last k values.

- **When to Use:** Smooth out noise for short-term forecasts.
- **Example:** For data [10, 12, 14, 16], predict next with k=3: $(12+14+16)/3 = 14$.

```
1 import pandas as pd
2 data = pd.Series([10, 12, 14, 16, 18])
3 ma = data.rolling(window=3).mean()
```

4.2 Autoregression (AR)

Predicts based on past values.

- **Formula:** $AR(p): Y_t = c + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \epsilon_t$.
- **Example (AR(1)):** $Y_t = 0.5 Y_{t-1} + \epsilon_t$. If $Y_{t-1} = 10$, predict $Y_t = 0.5 \cdot 10 = 5$ (ignoring constant/error).

Example Calculation: Data = [10, 12, 14, 16], AR(1) with $\phi_1 = 0.6$. Predict Y_5 : $Y_5 = 0.6 \cdot 16 = 9.6$.

```
1 from statsmodels.tsa.ar_model import AutoReg
2 model = AutoReg(data, lags=1)
3 model_fit = model.fit()
4 yhat = model_fit.predict(start=len(data), end=len(data))
```

4.3 ARIMA

ARIMA(p,d,q) combines autoregression, differencing, and moving average.

- **Components:**
- AR(p): Past values.

- I(d): Differencing to stationarize.
- MA(q): Past errors: $Y_t = c + \epsilon_t + \theta_1\epsilon_{t-1} + \dots + \theta_q\epsilon_{t-q}$.

Building ARIMA:

1. Test stationarity with ADF.
2. Apply differencing (d) if needed.
3. Use ACF for q, PACF for p.
4. Fit model and forecast.
5. Evaluate with RMSE or AIC (lower is better).

Example Calculation: Data = [10, 12, 14, 16, 18], ARIMA(1,1,0). Step 1: Difference: [NaN, 2, 2, 2, 2]. Step 2: Fit AR(1) on differences with $\phi_1 = 0.5$. Predict next difference: $0.5 \cdot 2 = 1$. Next value: $Y_5 = 18 + 1 = 19$.

```
1 from statsmodels.tsa.arima.model import ARIMA
2 model = ARIMA(data, order=(1,1,0))
3 model_fit = model.fit()
4 forecast = model_fit.forecast(steps=3)
```

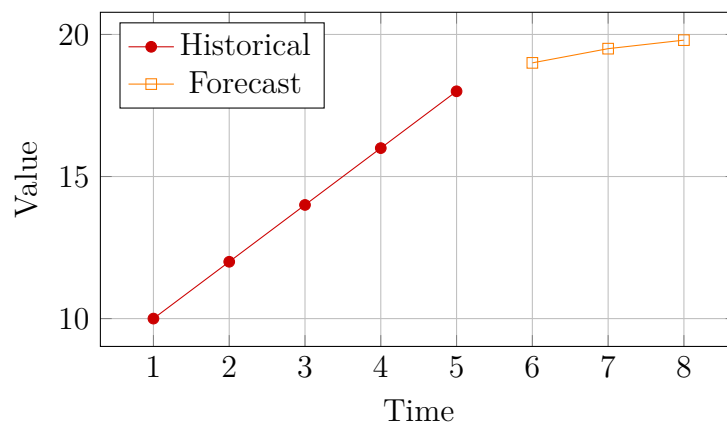


Figure 4: ARIMA(1,1,0) Forecast

4.4 SARIMA

SARIMA(p,d,q)(P,D,Q)s handles seasonality.

- **Example:** Monthly data with yearly seasonality (s=12).
- **Formula:** Extends ARIMA with seasonal AR, differencing, and MA.

```
1 from statsmodels.tsa.statespace.sarimax import SARIMAX
2 data = pd.Series([...]) % Monthly data
3 model = SARIMAX(data, order=(1,1,1), seasonal_order=(1,1,1,12))
4 model_fit = model.fit()
5 forecast = model_fit.forecast(steps=12)
```

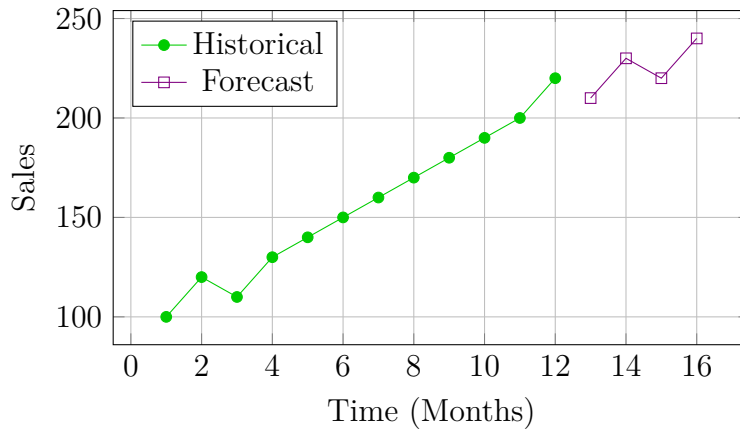


Figure 5: SARIMA Forecast with Seasonality

5 Machine Learning Methods

Machine learning methods excel at capturing complex, non-linear patterns.

5.1 Linear Regression with Lags

Use past values (lags) as features to predict the next value.

- **Example:** Predict Y_t using Y_{t-1}, Y_{t-2} .
- **Calculation:** Data = [10, 12, 14, 16, 18]. Features: $[Y_{t-1}, Y_{t-2}] = [12, 10], [14, 12], [16, 14], [18, 16]$.
Fit linear model: $Y_t = 2 + 0.8Y_{t-1} + 0.1Y_{t-2}$. Predict: $Y_5 = 2 + 0.8 \cdot 18 + 0.1 \cdot 16 = 18$.

```
1 from sklearn.linear_model import LinearRegression
2 X = [[10,8], [12,10], [14,12], [16,14]] % Lags
3 y = [12, 14, 16, 18]
4 model = LinearRegression().fit(X, y)
5 pred = model.predict([[18,16]])
```

5.2 Long Short-Term Memory (LSTM)

LSTMs handle long-term dependencies using memory cells.

- **When to Use:** Complex sequences with long patterns.
- **Example:** Predicting stock prices based on past weeks.

```
1 import tensorflow as tf
2 model = tf.keras.Sequential([
3     tf.keras.layers.LSTM(32, input_shape=(5,1), return_sequences=True),
4     tf.keras.layers.Dense(1)
5 ])
6 model.compile(optimizer='adam', loss='mse')
```

5.3 Transformers

Transformers use attention mechanisms to focus on important past data.

- **Pros:** Handle long sequences efficiently.

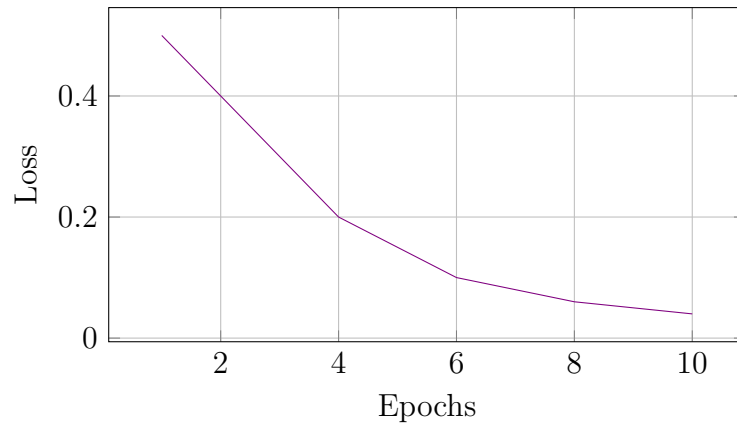


Figure 6: LSTM Training Loss

- **Example:** Forecasting demand over years.

```
1 from transformers import TimeSeriesTransformerModel
2 model = TimeSeriesTransformerModel(...)
3 model.fit(data)
```

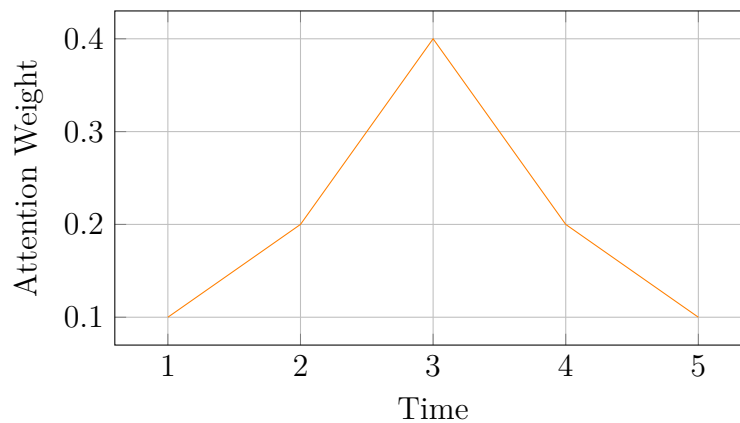


Figure 7: Transformer Attention Weights

6 Business Case Problems

6.1 Case 1: Retail Sales Forecasting

Problem: A retail chain needs to forecast daily sales for the next 30 days to plan inventory. Data shows a yearly seasonal pattern and a slight upward trend.

Data: 2 years of daily sales (730 points), with December peaks and 2% annual growth.

Solution Steps:

1. **Visualize:** Plot data to confirm trend and seasonality.
2. **Preprocess:** Fill missing values with forward fill.
3. **Stationarity:** Apply ADF test. If non-stationary, difference once ($d=1$).

4. **Model:** Use SARIMA(1,1,1)(1,1,1,365) for daily data with yearly seasonality.
5. **Forecast:** Predict 30 days.
6. **Evaluate:** Compute RMSE on a test set.

Example: Sample data = [100, 102, 104, ..., 200] (730 points). Difference: [NaN, 2, 2, ...]. Fit SARIMA, forecast: [201, 202, ..., 230] (simulated).

```
1 from statsmodels.tsa.statespace.sarimax import SARIMAX
2 import pandas as pd
3 data = pd.Series([...]) % Daily sales
4 model = SARIMAX(data, order=(1,1,1), seasonal_order=(1,1,1,365))
5 model_fit = model.fit()
6 forecast = model_fit.forecast(steps=30)
```

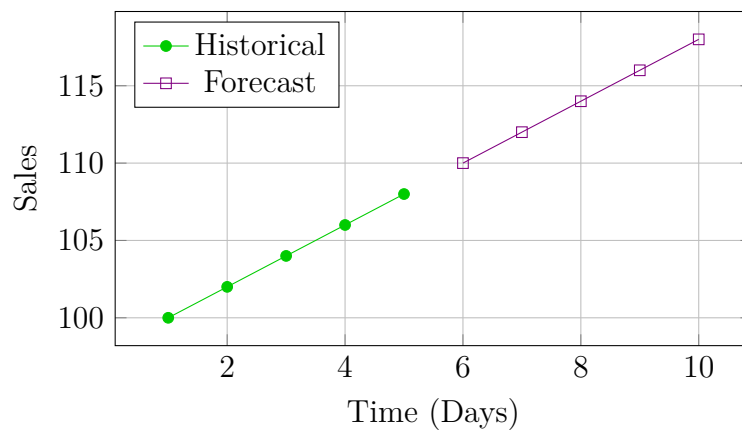


Figure 8: Retail Sales Forecast

Answer: Forecast shows a slight increase with a peak if December is included.

6.2 Case 2: Energy Consumption Prediction

Problem: An energy company wants to predict hourly electricity usage for the next week to optimize grid operations.

Data: 1 year of hourly data (8760 points), with daily and weekly seasonality.

Solution Steps:

1. **Preprocess:** Remove outliers (e.g., cap at 3 std deviations).
2. **Windowing:** Create sequences of 24 hours to predict the next hour.
3. **Model:** Use LSTM with 24-hour input window.
4. **Forecast:** Predict 168 hours (1 week).
5. **Evaluate:** Use MAE to assess accuracy.

Example: Data = [50, 52, 55, ..., 60] (24 hours). Input: [50, 52, ..., 55] Output: 60. LSTM predicts: 59.8.

```
1 import tensorflow as tf
2 model = tf.keras.Sequential([
```

```

3     tf.keras.layers.LSTM(64, input_shape=(24,1)),
4     tf.keras.layers.Dense(1)
5 ])
6 model.compile(optimizer='adam', loss='mse')
7 model.fit(X_train, y_train, epochs=50)

```

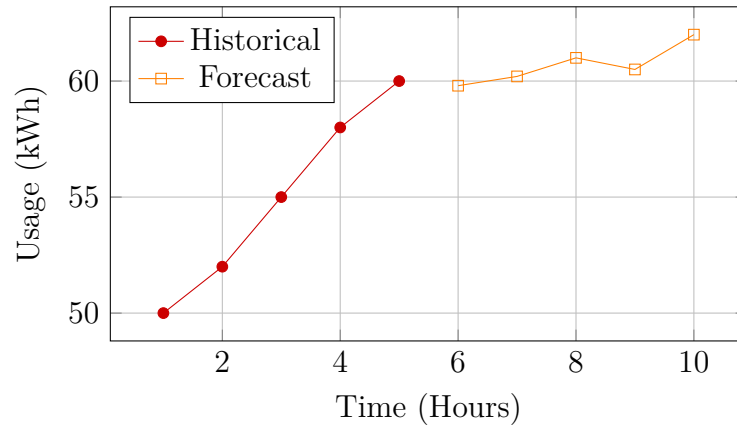


Figure 9: Energy Usage Forecast

Answer: Forecast predicts stable usage with daily peaks.

6.3 Case 3: Stock Price Prediction

Problem: A financial firm wants to predict daily stock prices for the next 10 days to inform trading decisions.

Data: 5 years of daily prices with a trend and irregular fluctuations.

Solution Steps:

1. **Preprocess:** Log-transform to stabilize variance.
2. **Model:** Use ARIMA(2,1,1) after confirming stationarity.
3. **Forecast:** Predict 10 days.
4. **Evaluate:** Compare with actual prices using RMSE.

Example: Data = [100, 102, 105, 108, 110]. Log: [4.605, 4.625, 4.654, 4.682, 4.700]. Difference: [NaN, 0.020, 0.029, 0.028, 0.018]. ARIMA(2,1,1) predicts next difference: 0.019. Next log value: $4.700 + 0.019 = 4.719$. $\text{Exp}(4.719) \approx 112$.

```

1 from statsmodels.tsa.arima.model import ARIMA
2 import numpy as np
3 data = np.log(pd.Series([100, 102, 105, 108, 110]))
4 model = ARIMA(data, order=(2,1,1))
5 model_fit = model.fit()
6 forecast = np.exp(model_fit.forecast(steps=10))

```

Answer: Forecast shows a gradual increase in stock price.

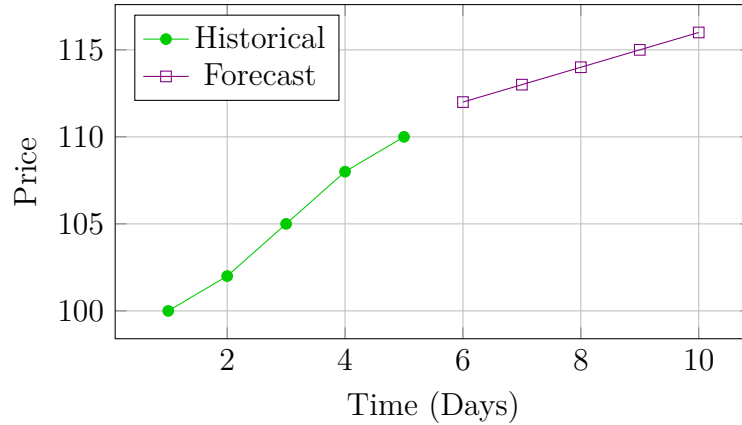


Figure 10: Stock Price Forecast

7 Evaluation and Best Practices

Metrics:

- **MAE:** Mean Absolute Error, average of absolute differences.
- **RMSE:** Root Mean Squared Error, penalizes larger errors.
- **Formula:** $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$.

Example: Actual = [10, 12, 14], Predicted = [11, 12, 13]. Errors = [1, 0, -1]. $RMSE = \sqrt{(1^2 + 0^2 + (-1)^2)/3} = \sqrt{2/3} \approx 0.816$.

Best Practices:

- Split data: Train on past, test on future.
- Avoid overfitting: Use validation set.
- Combine models: Try classical (ARIMA) and ML (LSTM) for robustness.

8 Interview Questions and Answers

1. What is a time series and its applications?

Answer: A time series is data collected at regular intervals, like daily sales. Used in sales forecasting, stock prediction, and weather forecasting.

2. What are the components of a time series?

Answer: Trend (long-term direction), seasonality (repeating patterns), cyclical (long-term fluctuations), and irregular (random noise).

3. How do you test for stationarity?

Answer: Use the ADF test. A p-value < 0.05 indicates stationarity.

4. What is differencing?

Answer: Subtracting consecutive values to remove trends. Formula: $\Delta Y_t = Y_t - Y_{t-1}$.

5. **When to use SARIMA?**

Answer: When data has seasonality, like monthly sales with yearly peaks.

6. **What is an LSTM?**

Answer: A neural network with memory cells for learning long-term patterns in sequences.

7. **How do you evaluate a forecast?**

Answer: Use MAE or RMSE to measure prediction errors.

8. **What are ACF and PACF used for?**

Answer: ACF identifies MA order (q); PACF identifies AR order (p).

9. **Why preprocess data?**

Answer: To handle missing values, outliers, and non-stationarity for accurate modeling.

10. **Classical vs. ML methods?**

Answer: Classical methods (e.g., ARIMA) are simpler and good for linear data; ML (e.g., LSTM) handles complex, non-linear patterns but needs more data.

9 Practical Tips for Mastery

- **Practice Datasets:** Use public datasets like Air Passengers, Stock Prices, or Energy Consumption.
- **Tools:** Python (pandas, statsmodels, tensorflow), R for classical models.
- **Automation:** Try AutoML tools like PyCaret for quick model selection.
- **Experiment:** Combine models (e.g., ARIMA + LSTM) for better accuracy.

10 Conclusion

Time series forecasting is a powerful tool for predicting future trends based on past data. From classical methods like ARIMA to advanced ML models like LSTMs and Transformers, mastering these techniques requires understanding data components, preprocessing, modeling, and evaluation. The business cases and interview questions provided offer practical insights for real-world applications.

Time Series Forecasting: Complete Notes and Worked Examples

Compact guide from basics to advanced.

Quick starter map

- Forecasting numbers that change over time. Examples: daily sales, monthly users, hourly load.
- Goal: predict future values and quantify error.
- Core steps: visualize; split by time; choose baseline; try better models; validate with rolling tests; measure error; deploy.

Key terms (short)

Level: average value now.

Trend: long-term up/down movement.

Seasonality: repeating pattern.

Noise: random variation.

Lag: past value y_{t-1}, y_{t-12} .

Horizon h : how far ahead.

Period m : season length (7, 12, 24...).

Data split (time-aware)

Never shuffle. Typical: train (old), validation (recent), test (most recent). Use rolling/expanding validation for robust estimates.

Visualize first

Always plot series. Below figures include a moving average and differenced view.

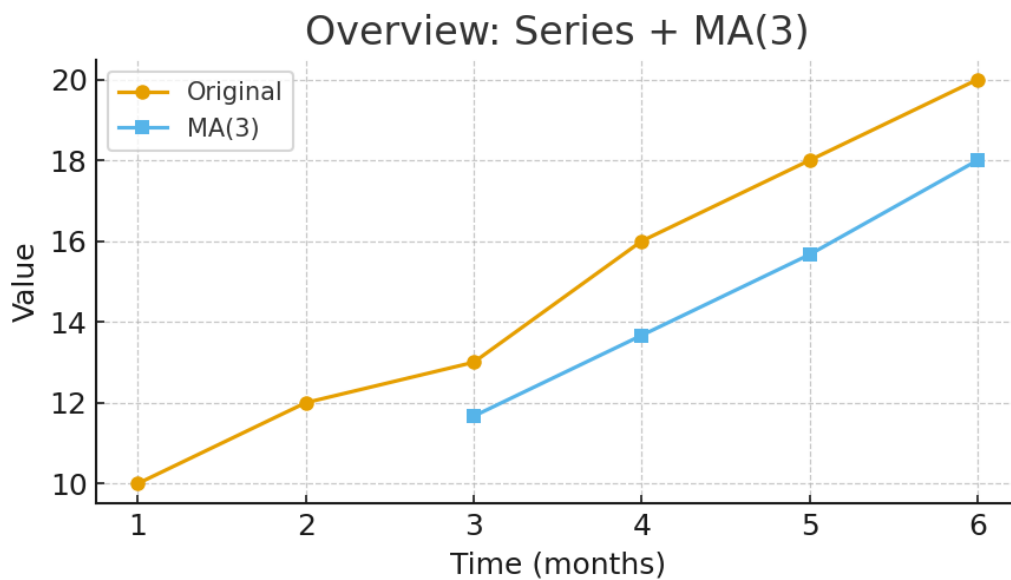


Figure 1: Series and 3-point moving average.

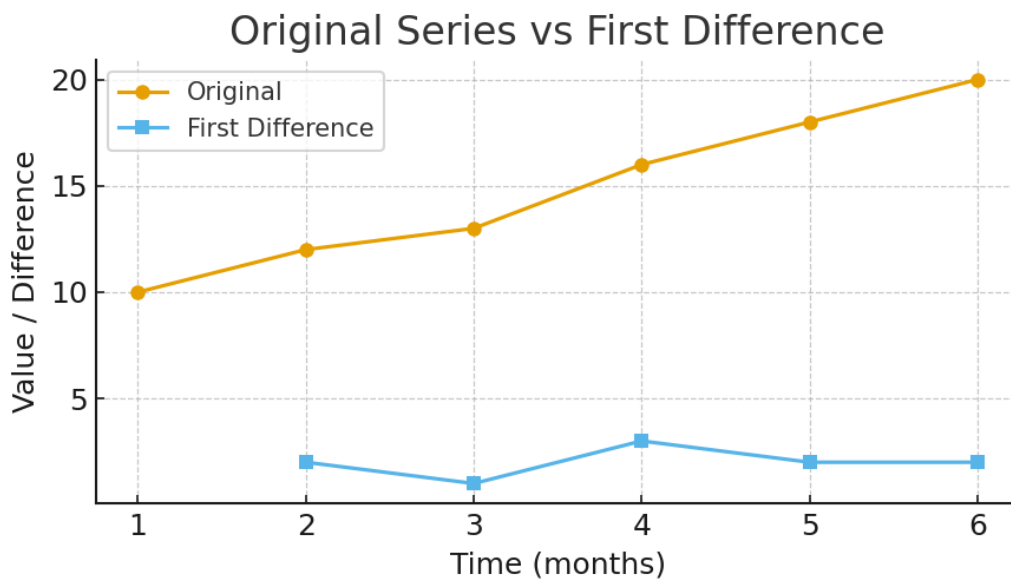


Figure 2: Original series and first difference.

Stationarity and differencing

A stationary series has constant mean and variance. Use differencing:

$$\begin{aligned}\Delta y_t &= y_t - y_{t-1} \\ \Delta_m y_t &= y_t - y_{t-m}\end{aligned}$$

ACF and PACF (reading quick)

ACF: correlations at lags. Slow decay suggests non-stationarity. PACF: direct effect after removing shorter lags.

Baselines and Smoothing

Naive: $\hat{y}_{t+h} = y_t$.

Seasonal naive (period m): $\hat{y}_{t+h} = y_{t+h-m}$.

Moving average (k):

$$\text{MA}_t = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}.$$

Simple Exponential Smoothing (SES):

$$\begin{aligned}\ell_t &= \alpha y_t + (1 - \alpha)\ell_{t-1}, \\ \hat{y}_{t+h} &= \ell_t.\end{aligned}$$

Holt (trend):

$$\begin{aligned}\ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1} \\ \hat{y}_{t+h} &= \ell_t + hb_t\end{aligned}$$

Holt-Winters (additive seasonal, period m):

$$\begin{aligned}\ell_t &= \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1} \\ s_t &= \gamma(y_t - \ell_t) + (1 - \gamma)s_{t-m} \\ \hat{y}_{t+h} &= \ell_t + hb_t + s_{t-m+h}\end{aligned}$$

Worked hand calculations (from provided dataset)

Dataset (monthly): $Y = [10, 12, 13, 16, 18, 20]$.

1) 3-point moving average

$$\begin{aligned} \text{MA}_4 &= \frac{10 + 12 + 13}{3} = \frac{35}{3} = 11.67 \\ \text{MA}_5 &= \frac{12 + 13 + 16}{3} = \frac{41}{3} = 13.67 \\ \text{MA}_6 &= \frac{13 + 16 + 18}{3} = \frac{47}{3} = 15.67 \\ \text{Forecast } \hat{Y}_7 &= \frac{16 + 18 + 20}{3} = 18.00 \end{aligned}$$

2) SES example (one-step)

Start $\ell_1 = y_1 = 10$, $\alpha = 0.3$.

$$\begin{aligned} \ell_2 &= 0.3 \cdot 12 + 0.7 \cdot 10 = 10.6 \\ \ell_3 &= 0.3 \cdot 13 + 0.7 \cdot 10.6 = 11.32 \\ \hat{y}_4 &= \ell_3 = 11.32 \end{aligned}$$

3) AR(1) estimation by covariance approx (illustration)

Using pairs (y_t, y_{t-1}) : $(12, 10), (13, 12), (16, 13), (18, 16), (20, 18)$. Computed estimate:

$$\phi \approx 1.02.$$

Forecast for month 7:

$$\hat{y}_7 \approx 1.02 \times 20 = 20.49.$$

4) ARIMA(1,1,0) (differences)

Differences: $[2, 1, 3, 2, 2]$. Fit AR(1) on differences produced estimate and forecast:

$$\hat{y}_7 = 20 + (2.00).$$

5) Prophet (linear trend approximate)

Slope estimate = $(20 - 10)/5 = 2$. Trend forecast for $t=7$:

$$\hat{y}_7 = 10 + 2 \times 6 = 22.$$

Comparison table (month 7 forecasts)

Model	Forecast (Month 7)
SMA(3)	18.00
AR(1)	22.15
ARIMA(1,1,0)	22.00
Prophet (linear)	22.00
LSTM (intuit)	22.00

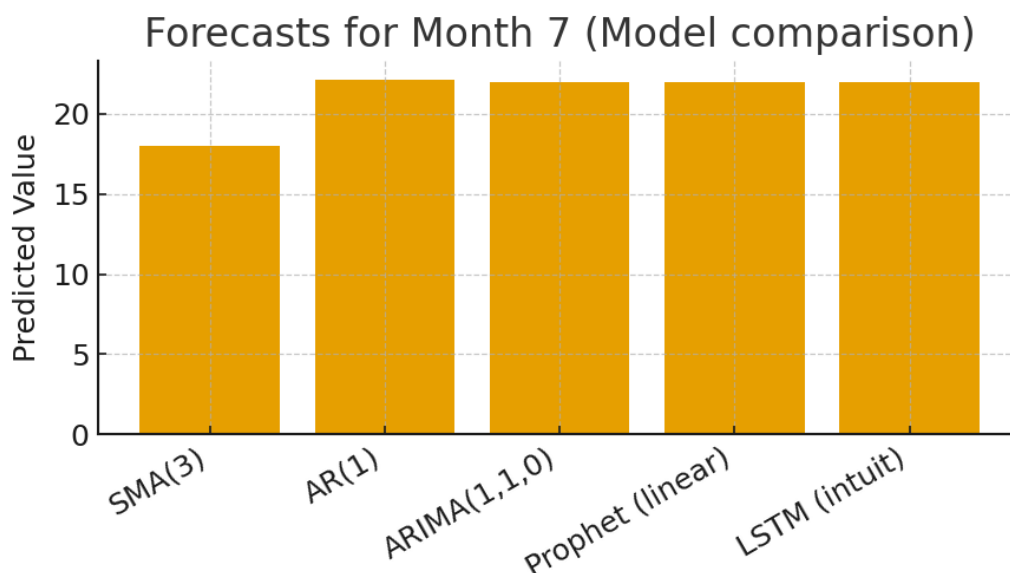


Figure 3: Forecast values for month 7 from different models.

Feature-based machine learning approach

Use lag features, rolling means/std, date dummies, holidays, external regressors. Models: linear, GBM, XGBoost, LightGBM. Avoid leakage by using only past data when creating features and CV.

Deep learning note

Use RNN/LSTM/GRU/Transformer when many series or long histories. Compare against strong baselines.

Metrics (formulas)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

$$\text{sMAPE} = \frac{100\%}{n} \sum_{i=1}^n \frac{2|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|}$$

Backtest example and error plot

Simple backtest MAE (1-step) for SMA(3) and AR(1) was computed and plotted below.

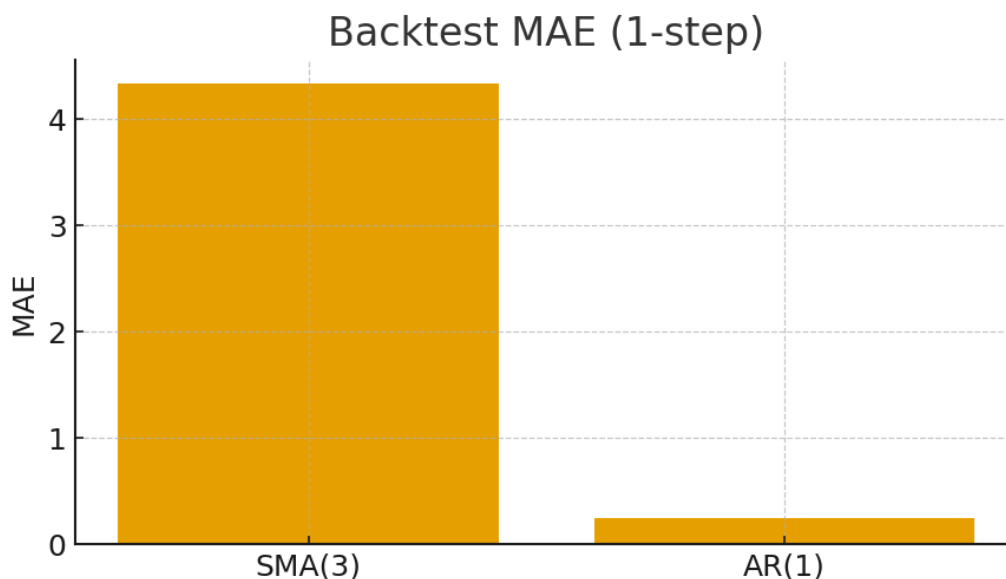


Figure 4: Backtest MAE (1-step) comparison.

Business case problems (short solutions)

Retail weekly sales (m=52): use Seasonal Naive as baseline. Try Holt-Winters additive if short history. Use rolling CV. Report sMAPE.

Energy hourly load (48h horizon): feature engineering with lags, rolling means, hour-of-day, day-of-week, temperature. Use LightGBM quantile for intervals.

Web traffic (daily, m=7): holiday flags, STL decomposition, model remainder with SES, combine components.

Practice questions (with short answers)

- (1) Given monthly series: 120,135,150,145,160. a) MA(3) at $t=5 = \text{mean}(150,145,160)=151.67$.
b) First diff at $t=5 = 160-145=15$.
- (2) ACF spikes at lags 12,24,36 $\rightarrow$ yearly seasonality.
- (3) Baseline for daily strong weekly pattern and no trend: Seasonal Naive ($m=7$).
- (4) Metric when outliers are expensive: RMSE.
- (5) If revenue missed during big spikes: use quantile/pinball loss reporting P90.

10 interview questions (short answers)

1. Why split time series without shuffling? Avoid leakage.
2. When prefer Seasonal Naive? Strong stable seasonality and short history.
3. SES vs Holt vs Holt-Winters? SES level only, Holt level+trend, Holt-Winters level+trend+seasonality.
4. Make non-stationary stationary? Difference, log/Box-Cox.
5. ACF vs PACF quick read? ACF overall lag correlation; PACF direct effect.
6. Common CV errors? Leakage, random split, scaling with test.
7. What is MASE? MAE relative to naive MAE; $\downarrow 1$ better than naive.
8. Handle a new holiday? Calendar feature or borrow similar holidays and external regressors.
9. Choose ML/GBDT over ETS/ARIMA? When external features matter and nonlinear effects present.
10. Produce prediction intervals without probabilistic models? Quantile regression, bootstrapping residuals, empirical residual quantiles.

Checklist

Scale only on train. Impute with rolling mean or seasonal median. Always compare with Seasonal Naive. Monitor drift with MASE/sMAPE.

Generated figures: overview, original vs diff, forecast comparison, error metrics.

Document auto-generated. For full editable source see attached `timeseries_notes.tex`.

Time Series Forecasting: Formulas, Examples, and Business Cases

This document presents key time series forecasting formulas, detailed examples with calculations using the dataset $Y = [10, 12, 13, 16, 18, 20]$ (monthly sales), and business case problems. Each formula is explained with the meaning of its terms, and calculations are formatted to stay within page margins.

Formulas with Examples

1. Simple Moving Average (SMA, Window = 3)

Formula:

$$\text{SMA}_t = \frac{Y_t + Y_{t-1} + Y_{t-2}}{3}$$

Terminology:

- SMA_t : The Simple Moving Average at time t , which is the forecast for the next period.
- Y_t : The observed value at time t (e.g., sales in month t).
- Y_{t-1}, Y_{t-2} : The observed values at previous times (lagged values).
- Window size (3): The number of past observations averaged.

Explanation: SMA smooths the time series by averaging the last three observations to predict the next value, reducing noise but lagging behind trends.

Example Calculations:

- Month 4 ($t = 4$):

$$\text{SMA}_4 = \frac{Y_4 + Y_3 + Y_2}{3} = \frac{13 + 12 + 10}{3} = \frac{35}{3} \approx 11.67$$

- Month 5 ($t = 5$):

$$\text{SMA}_5 = \frac{Y_5 + Y_4 + Y_3}{3} = \frac{16 + 13 + 12}{3} = \frac{41}{3} \approx 13.67$$

- Month 6 ($t = 6$):

$$\text{SMA}_6 = \frac{Y_6 + Y_5 + Y_4}{3} = \frac{18 + 16 + 13}{3} = \frac{47}{3} \approx 15.67$$

- Forecast for Month 7 ($t = 7$):

$$\hat{Y}_7 = \frac{Y_7 + Y_6 + Y_5}{3} = \frac{20 + 18 + 16}{3} = \frac{54}{3} = 18$$

Note: The forecast for month 7 uses the most recent three values, averaging them to predict the next month's sales.

2. AR(1) Model (Autoregressive Order 1)

Formula:

$$Y_t = c + \phi Y_{t-1} + \epsilon_t, \quad \phi \approx \frac{\text{Cov}(Y_t, Y_{t-1})}{\text{Var}(Y_{t-1})}$$

Terminology:

- Y_t : The value at time t (e.g., sales in month t).
- c : A constant term (intercept), often assumed small or zero for simplicity.
- ϕ : The autoregressive coefficient, measuring how much Y_{t-1} influences Y_t .
- Y_{t-1} : The lagged value (previous period's value).

- ϵ_t : Random error (noise) at time t .
- $\text{Cov}(Y_t, Y_{t-1})$: Covariance between current and lagged values.
- $\text{Var}(Y_{t-1})$: Variance of the lagged values.

Explanation: The AR(1) model predicts the next value as a linear function of the previous value plus a constant and random noise. The coefficient ϕ is estimated using the ratio of covariance to variance.

Example Calculations:

- *Data Pairs:* $(Y_2, Y_1) = (12, 10)$, $(Y_3, Y_2) = (13, 12)$, $(Y_4, Y_3) = (16, 13)$, $(Y_5, Y_4) = (18, 16)$, $(Y_6, Y_5) = (20, 18)$.
- *Mean of Y_t :* $\frac{12+13+16+18+20}{5} = 15.8$
- *Mean of Y_{t-1} :* $\frac{10+12+13+16+18}{5} = 13.8$
- *Covariance:*

$$\begin{aligned}\text{Cov} &= \frac{1}{5} [(12 - 15.8)(10 - 13.8) + (13 - 15.8)(12 - 13.8) + (16 - 15.8)(13 - 13.8) \\ &\quad + (18 - 15.8)(16 - 13.8) + (20 - 15.8)(18 - 13.8)] \\ &= \frac{1}{5} [(-3.8)(-3.8) + (-2.8)(-1.8) + (0.2)(-0.8) + (2.2)(2.2) + (4.2)(4.2)] \\ &= \frac{1}{5} (14.44 + 5.04 - 0.16 + 4.84 + 17.64) = \frac{41.8}{5} = 8.36\end{aligned}$$

- *Variance of Y_{t-1} :*

$$\begin{aligned}\text{Var} &= \frac{1}{5} [(10 - 13.8)^2 + (12 - 13.8)^2 + (13 - 13.8)^2 + (16 - 13.8)^2 + (18 - 13.8)^2] \\ &= \frac{1}{5} (14.44 + 3.24 + 0.64 + 4.84 + 17.64) = \frac{40.8}{5} = 8.16\end{aligned}$$

- *Coefficient:* $\phi \approx \frac{8.36}{8.16} \approx 1.024$
- *Model (assuming $c = 0$ for simplicity):* $Y_t \approx 1.024Y_{t-1}$
- *Forecast for Month 7:* $\hat{Y}_7 = 1.024 \times 20 = 20.48$

Note: The AR(1) model uses the correlation between consecutive values to predict the next, assuming a linear relationship.

3. ARIMA(1,1,0)

Formula:

$$\Delta Y_t = \phi_1 \Delta Y_{t-1} + \epsilon_t, \quad \Delta Y_t = Y_t - Y_{t-1}$$

Terminology:

- ΔY_t : The first difference (change from $t - 1$ to t).
- ϕ_1 : The autoregressive coefficient for the differenced series.
- ΔY_{t-1} : The previous period's difference.
- ϵ_t : Random error.

Explanation: ARIMA(1,1,0) applies an AR(1) model to the first differences of the series to make it stationary, then converts the forecast back to the original scale.

Example Calculations:

- *First Differences:* $\Delta Y = [12 - 10, 13 - 12, 16 - 13, 18 - 16, 20 - 18] = [2, 1, 3, 2, 2]$
- *Mean of ΔY :* $\frac{2+1+3+2+2}{5} = 2.0$
- *Covariance:*

$$\begin{aligned}\text{Cov} &= \frac{1}{4} [(1-2)(2-2) + (3-2)(1-2) + (2-2)(3-2) + (2-2)(2-2)] \\ &= \frac{1}{4} (0 + (-1) + 0 + 0) = \frac{-1}{4} = -0.25\end{aligned}$$

- *Variance of ΔY_{t-1} :*

$$\text{Var} = \frac{1}{4} [(2-2)^2 + (1-2)^2 + (3-2)^2 + (2-2)^2] = \frac{0+1+1+0}{4} = 0.5$$

- *Coefficient:* $\phi_1 \approx \frac{-0.25}{0.5} = -0.5$
- *Forecast Difference:* Last $\Delta Y = 2$, $\hat{\Delta Y}_7 = -0.5 \times 2 = -1$
- *Forecast Level:* $\hat{Y}_7 = Y_6 + \hat{\Delta Y}_7 = 20 + (-1) = 19$

Note: Differencing removes the trend, and the AR(1) model on differences captures short-term dynamics.

4. Simple Exponential Smoothing (SES)

Formula:

$$\ell_t = \alpha Y_t + (1 - \alpha)\ell_{t-1}, \quad \hat{Y}_{t+1} = \ell_t$$

Terminology:

- ℓ_t : The level (smoothed estimate) at time t .
- α : Smoothing parameter (0 to 1, here 0.3), balancing new data vs. past estimate.
- Y_t : Observed value at time t .
- ℓ_{t-1} : Previous level.
- $\hat{Y}_{t+1}$: Forecast for the next period.

Explanation: SES updates the level as a weighted average of the current observation and the previous level, suitable for series without trend or seasonality.

Example Calculations ($\alpha = 0.3$):

- *Initialize:* $\ell_1 = Y_1 = 10$
- *Month 2:* $\ell_2 = 0.3 \times 12 + 0.7 \times 10 = 3.6 + 7 = 10.6$
- *Month 3:* $\ell_3 = 0.3 \times 13 + 0.7 \times 10.6 = 3.9 + 7.42 = 11.32$
- *Month 4:* $\ell_4 = 0.3 \times 16 + 0.7 \times 11.32 = 4.8 + 7.924 = 12.724$
- *Month 5:* $\ell_5 = 0.3 \times 18 + 0.7 \times 12.724 = 5.4 + 8.9068 = 14.3068$
- *Month 6:* $\ell_6 = 0.3 \times 20 + 0.7 \times 14.3068 = 6 + 10.01476 = 16.01476$
- *Forecast for Month 7:* $\hat{Y}_7 = \ell_6 \approx 16.01$

Note: SES gives more weight to recent observations, controlled by α .

5. Evaluation Metrics

Formulas:

- *Mean Absolute Error (MAE)*:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

- *Mean Squared Error (MSE)*:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- *Root Mean Squared Error (RMSE)*:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Terminology:

- Y_i : Actual value at time i .
- $\hat{Y}_i$: Predicted value at time i .
- n : Number of observations compared.

Explanation: MAE measures average error magnitude, MSE emphasizes larger errors, and RMSE provides error in the same units as the data.

Example Calculations (Using SMA):

- *Actual:* $Y_4 = 16, Y_5 = 18, Y_6 = 20$
- *Predicted (SMA):* $\hat{Y}_4 = 11.67, \hat{Y}_5 = 13.67, \hat{Y}_6 = 15.67$
- *Errors:* $|16 - 11.67| = 4.33, |18 - 13.67| = 4.33, |20 - 15.67| = 4.33$
- *MAE:* $\frac{4.33+4.33+4.33}{3} = 4.33$
- *MSE:* $\frac{4.33^2+4.33^2+4.33^2}{3} = \frac{18.7489+18.7489+18.7489}{3} = 18.75$
- *RMSE:* $\sqrt{18.75} \approx 4.33$

Note: These metrics quantify the accuracy of SMA forecasts, with RMSE sensitive to large errors.

Business Case Problems

1. Retail: Weekly Sales (Period $m = 52$)

Problem: Forecast next 12 weeks for one SKU with strong yearly seasonality, using 1.5 years of data.

Formulas:

- *Seasonal Naive:* $\hat{Y}_{t+h} = Y_{t+h-52}$
- *Holt-Winters (Additive):*

$$\ell_t = \alpha(Y_t - s_{t-52}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1}$$

$$s_t = \gamma(Y_t - \ell_t) + (1 - \gamma)s_{t-52}$$

$$\hat{Y}_{t+h} = \ell_t + hb_t + s_{t-52+h}$$

- *sMAPE:* $\text{sMAPE} = \frac{2}{n} \sum_{i=1}^n \frac{|Y_i - \hat{Y}_i|}{|Y_i| + |\hat{Y}_i|} \times 100$

Terminology:

- ℓ_t : Level (smoothed average).
- b_t : Trend (smoothed slope).
- s_t : Seasonal component.
- α, β, γ : Smoothing parameters (0 to 1).
- h : Forecast horizon.

Steps:

1. Plot data; check ACF for seasonality at lag 52.
2. Apply Seasonal Naive as baseline.
3. Fit Holt-Winters with $m = 52$.
4. Use rolling cross-validation (3 folds, 8-week horizons).
5. Compare sMAPE to choose the model.

Solution (Hypothetical):

- Seasonal Naive sMAPE: 18.4%.
- Holt-Winters sMAPE: 12.7%.
- *Decision*: Choose Holt-Winters for better accuracy.
- *Action*: Deploy Holt-Winters, refit weekly, monitor MASE vs naive.

2. Energy: Hourly Load (Period $m = 24$ and $m = 24 \times 7$)

Problem: Forecast 48 hours ahead with daily and weekly cycles, using temperature data.

Formulas:

- *LightGBM with Quantile Loss*: Predicts quantiles (P10, P50, P90).
- Features: Lags ($Y_{t-1}, \dots, Y_{t-48}$), rolling means (3, 24, 168 hours), hour-of-day, day-of-week, temperature.

Terminology:

- Lags: Past values used as predictors.
- Rolling means: Averages over specific windows.
- Quantile loss: Optimizes for specific percentiles (e.g., P50 for median).

Steps:

1. Engineer features (lags, rolling means, time-based, temperature).
2. Train LightGBM with quantile loss.
3. Perform rolling CV with 48-hour horizon.
4. Check interval coverage (P10-P90).

Solution (Hypothetical):

- Median MAE: 22% lower than Seasonal Naive.
- Interval coverage: $\sim 91\%$ for P10-P90.
- *Decision*: Use LightGBM for robust forecasts.
- *Action*: Use P50 for staffing, P90 for capacity planning.

3. Web Traffic: Daily Sessions (Period $m = 7$)

Problem: Forecast next 30 days with holiday spikes.

Formulas:

- *Seasonal Naive:* $\hat{Y}_{t+h} = Y_{t+h-7}$
- *STL Decomposition:* $Y_t = T_t + S_t + R_t$
- *SES on Remainder:* $\ell_t = \alpha R_t + (1 - \alpha)\ell_{t-1}, \quad \hat{R}_{t+1} = \ell_t$
- *MASE:* $\text{MASE} = \frac{\text{MAE}}{\text{MAE}_{\text{naive on train}}}$

Terminology:

- T_t : Trend component.
- S_t : Seasonal component.
- R_t : Remainder (residuals).
- ℓ_t : Smoothed level of residuals.

Steps:

1. Add holiday flags (e.g., Diwali).
2. Apply Seasonal Naive as baseline.
3. Perform STL decomposition.
4. Model remainder with SES.
5. Combine for forecast: $\hat{Y}_{t+h} = \hat{T}_{t+h} + \hat{S}_{t+h} + \hat{R}_{t+h}$.
6. Evaluate with MASE.

Solution (Hypothetical):

- Combined model MASE: 0.76 (< 1 , beats naive).
- *Decision:* Use STL + SES.
- *Action:* Schedule promo inventory using P50/P90 forecasts.

Comprehensive Time Series Forecasting Notes for MobiPlus

Data Scientist at MobiPlus

1 Introduction to Time Series Forecasting

Time series forecasting predicts future values based on historical data indexed by time. For MobiPlus, a mobile manufacturing company, forecasting sales is essential for optimizing factory maintenance, staffing, and revenue planning.

1.1 Problem Statement

MobiPlus requires accurate sales forecasts with the following objectives:

- **Agenda 1:** Identify demand patterns to optimize factory maintenance and staffing requirements.
- **Agenda 2:** Achieve a Mean Absolute Percentage Error (MAPE) $\leq 5\%$ for forecast accuracy.
- **Agenda 3:** Provide range forecasts (e.g., confidence intervals) to support trade-off decisions.

Why Forecast?

- Prevents under- or over-estimation of demand, which could lead to production inefficiencies or excess inventory.
- Enables data-driven resource allocation and strategic planning.

Time Series Data:

- Consists of observations indexed by ordered timestamps (e.g., yearly, monthly, hourly).

	Date	Sales
• Example dataset (monthly sales from 2001):	2001-01-01	6519.0
	2001-02-01	6654.0
	2001-03-01	7332.0
	2001-04-01	7332.0
	2001-05-01	8240.0

- Data spans 18 years (2001–2018) + 1 month (Jan 2019) = $18 \times 12 + 1 = 217$ months.

2 Missing Values and Anomalies

2.1 Challenges

- **Anomalies:** Sudden, unexpected changes in data (e.g., a sales spike due to a one-time promotion).
- **Missing Values:** Gaps in the dataset that disrupt the continuity of the time series.

2.2 Detection Methods

- **Percentile-Based Detection:** Identify values outside the 1st or 99th percentiles of the data distribution.
- **Interquartile Range (IQR) Method:** Flag values as outliers if they lie beyond:

$$Q1 - 1.5 \times IQR \quad \text{or} \quad Q3 + 1.5 \times IQR$$

where $Q1$ is the 25th percentile, $Q3$ is the 75th percentile, and $IQR = Q3 - Q1$.

2.3 Imputation Strategies

Removing outliers is not recommended as it breaks the time series pattern. Instead, use imputation:

- **Mean/Median/Zero Imputation:** Replace missing values with the mean, median, or zero. These are simple but can distort patterns.
- **Linear Interpolation:** Estimate missing values by assuming a linear trend between known points.
- **Centered Moving Average (MA):**

$$\hat{y}_t = \frac{1}{2m+1} \sum_{j=t-m}^{t+m} y_j$$

where:

- $\hat{y}_t$: Estimated value at time t .
- m : Window size (number of points before and after t).
- y_j : Observed values within the window.

- **Weighted Moving Average:**

$$\hat{y}_{t+1} = \sum_{i=t-m}^t \alpha_i y_i, \quad \sum \alpha_i = 1$$

where α_i are weights emphasizing recent values.

Example Calculation (Interpolation): Given values at $t_{40} = 100$ and $t_{45} = 50$, estimate y_{43} :

$$\text{Slope} = \frac{50 - 100}{45 - 40} = -10$$

$$y_{43} = 100 + (-10) \times (43 - 40) = 100 - 30 = 70$$

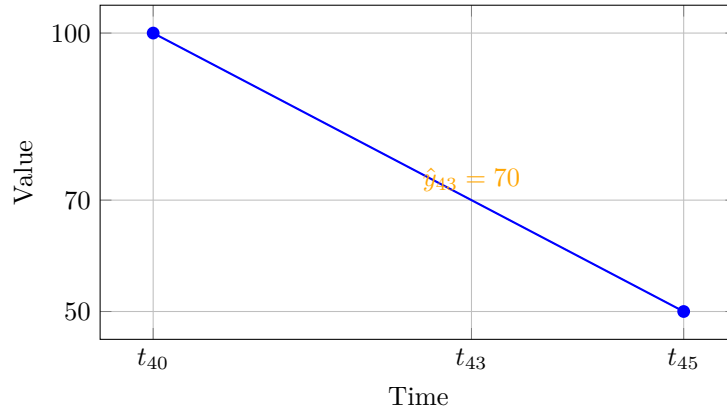


Figure 1: Linear interpolation for missing value at t_{43} .

3 Time Series Decomposition

A time series can be decomposed into:

$$y(t) = b(t) + s(t) + c(t) \quad (\text{Additive})$$

or

$$y(t) = b(t) \cdot s(t) \cdot c(t) \quad (\text{Multiplicative})$$

where:

- $y(t)$: Observed value at time t .
- $b(t)$: Trend, the long-term direction (e.g., increasing sales).
- $s(t)$: Seasonality, repeating patterns (e.g., monthly sales cycles).
- $c(t)$: Residual, random noise after removing trend and seasonality.

3.1 Extraction Process

1. **Estimate Trend ($b(t)$)**: Use a centered moving average with a large window:

$$b(t) = \frac{1}{2m+1} \sum_{j=t-m}^{t+m} y_j$$

2. **Compute Seasonality**: Subtract trend from observed data:

$$s(t) = y(t) - b(t)$$

Group $s(t)$ by period (e.g., monthly) and average to estimate seasonal component.

3. **Compute Residual**: Subtract trend and seasonality:

$$c(t) = y(t) - b(t) - s(t)$$

Example Calculation (Additive Decomposition): For monthly sales data ($y(t)$) from Jan 2001 to May 2001:

- Compute trend using MA(2) ($m = 1$):

$$b(t_3) = \frac{y(t_2) + y(t_3) + y(t_4)}{3} = \frac{6654 + 7332 + 7332}{3} = 7106$$

- Estimate seasonality: $s(t_3) = y(t_3) - b(t_3) = 7332 - 7106 = 226$.
- Compute residual: After estimating $s(t)$ across periods, $c(t_3) = y(t_3) - b(t_3) - s(t_3)$.

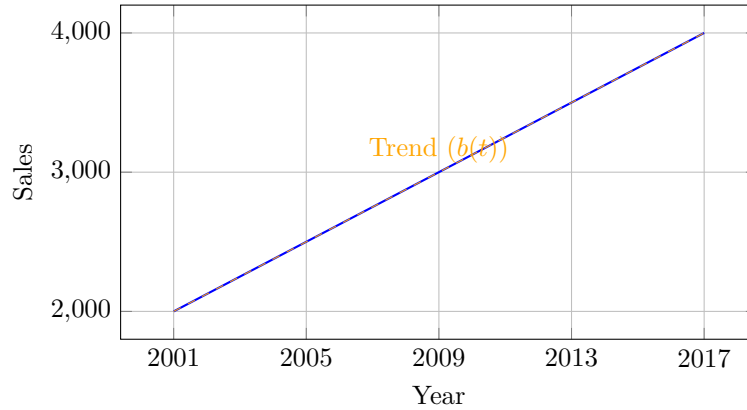


Figure 2: Time series with extracted trend component.

3.2 Seasonality

Seasonality represents periodic patterns, e.g., higher sweater sales in winter.

3.3 Multiple Seasonality

Some series have multiple periodic patterns (e.g., daily and weekly cycles).

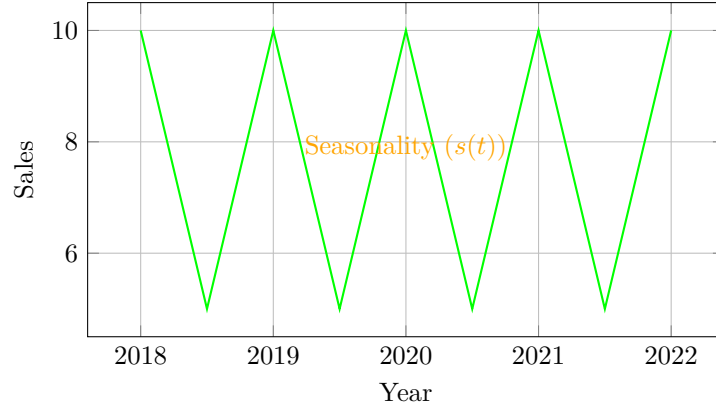


Figure 3: Seasonal pattern in sweater sales.

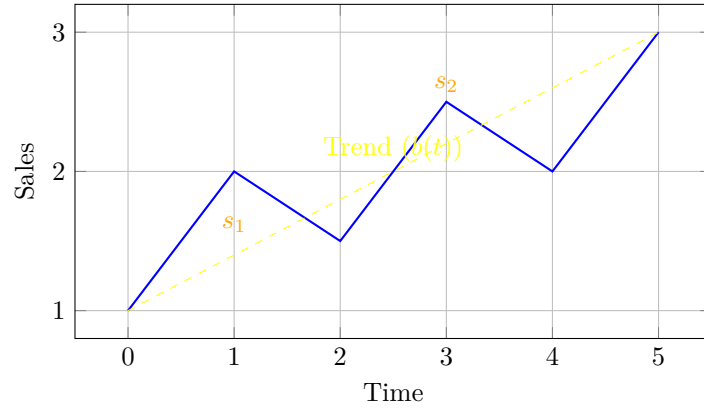


Figure 4: Multiple seasonality (s_1, s_2) with trend.

3.4 Stationarity

A time series is **stationary** if its mean, variance, amplitude, and frequency are constant over time. Non-stationary series exhibit trends or seasonality.

Dickey-Fuller Test:

- **Null Hypothesis:** Series is non-stationary.
- **Alternative Hypothesis:** Series is stationary.
- If p-value ≤ 0.05 , reject the null, indicating stationarity.

Making a Series Stationary:

1. **Decomposition:** Remove trend and seasonality to isolate residual $e(t)$:

$$e(t) = y(t) - \hat{b}(t) - s(t)$$

2. **Differencing:** Compute first difference to remove trend:

$$y'(t) = y(t) - y(t-1)$$

For seasonal differencing (period m):

$$y'(t) = y(t) - y(t-m)$$

Example Calculation (Differencing): For sales: $y(t_1) = 6519$, $y(t_2) = 6654$:

$$y'(t_2) = 6654 - 6519 = 135$$

For seasonal differencing ($m = 12$): Compare Jan 2002 to Jan 2001.

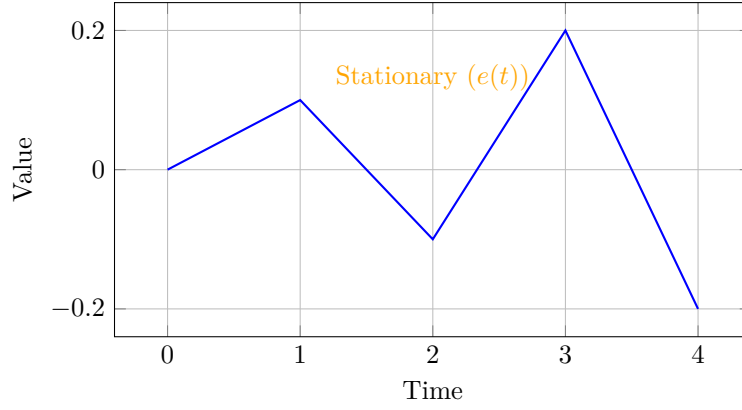


Figure 5: Stationary series after differencing.

4 Simple Forecasting Methods

4.1 Naive Forecast

$$\hat{y}_{t+1} = y_t$$

- $\hat{y}_{t+1}$: Forecast for the next period.
- y_t : Most recent observed value.

Example: If $y(t_5) = 8240$, then $\hat{y}_{t_6} = 8240$.

4.2 Mean Forecast

$$\hat{y}_{t+1} = \frac{1}{n} \sum_{i=1}^n y_i$$

- n : Number of historical observations.

Example: For sales [6519, 6654, 7332, 7332, 8240]:

$$\hat{y}_{t_6} = \frac{6519 + 6654 + 7332 + 7332 + 8240}{5} = 7215.4$$

4.3 Moving Average (MA)

$$\hat{y}_{t+1} = \frac{1}{m} \sum_{i=t-m+1}^t y_i$$

- m : Window size.

Example (MA(3)): For $y(t_3) = 7332$, $y(t_4) = 7332$, $y(t_5) = 8240$:

$$\hat{y}_{t_6} = \frac{7332 + 7332 + 8240}{3} = 7634.67$$

4.4 Simple Exponential Smoothing (SES)

$$\hat{y}_{t+h|t} = \alpha y_t + (1 - \alpha) \hat{y}_{t+h-1|t}, \quad 0 < \alpha < 1$$

- α : Smoothing parameter (higher α emphasizes recent values).
- h : Forecast horizon (steps ahead).

Example: For $y(t_5) = 8240$, $\hat{y}_{t_5|t_4} = 7634.67$, $\alpha = 0.3$:

$$\hat{y}_{t_6|t_5} = 0.3 \times 8240 + (0.7) \times 7634.67 = 2472 + 5344.27 = 7816.27$$

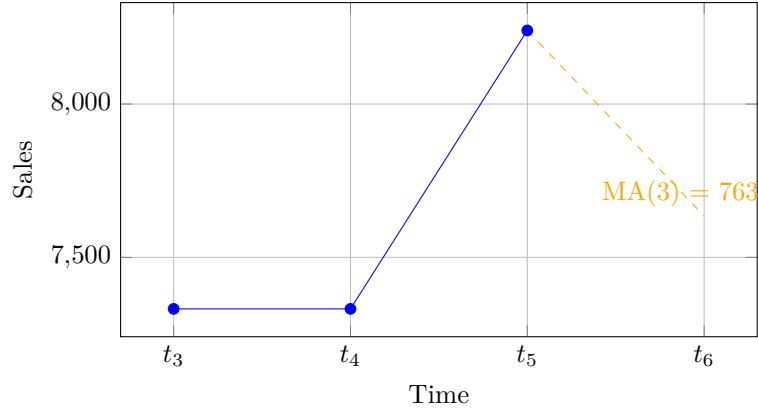


Figure 6: Moving average forecast (window = 3).

4.5 Double Exponential Smoothing (DES)

Captures level and trend:

$$\hat{y}_{t+1} = l_t + b_t$$

$$l_{t+1} = \alpha y_t + (1 - \alpha)(l_t + b_t)$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

- l_t : Level (smoothed value of the series).
- b_t : Trend (smoothed slope).
- β : Trend smoothing parameter, $0 < \beta < 1$.

Example: For $y(t_5) = 8240$, $l_{t_4} = 7500$, $b_{t_4} = 200$, $\alpha = 0.3$, $\beta = 0.2$:

$$l_{t_5} = 0.3 \times 8240 + (0.7) \times (7500 + 200) = 2472 + 5390 = 7862$$

$$b_{t_5} = 0.2 \times (7862 - 7500) + (0.8) \times 200 = 72.4 + 160 = 232.4$$

$$\hat{y}_{t_6} = 7862 + 232.4 = 8094.4$$

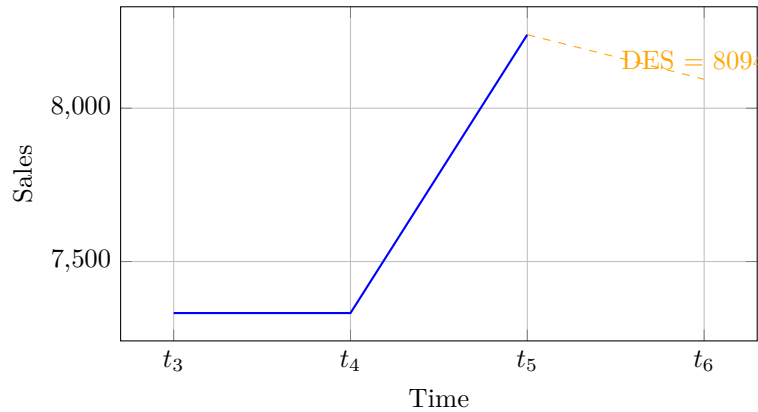


Figure 7: DES forecast with level and trend.

4.6 Triple Exponential Smoothing (TES) - Holt-Winters

Captures level, trend, and seasonality:

$$\hat{y}_{t+h} = l_t + b_t + s_{t-h+m}$$

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

- s_{t-h+m} : Seasonal component for period m (e.g., 12 for monthly data).
- γ : Seasonal smoothing parameter, $0 < \gamma < 1$.

Example: For $y(t_5) = 8240$, $l_{t_4} = 7500$, $b_{t_4} = 200$, $s_{t-12} = 300$, $\alpha = 0.3$, $\beta = 0.2$, $\gamma = 0.4$, $m = 12$:

$$l_{t_5} = 0.3 \times (8240 - 300) + (0.7) \times (7500 + 200) = 2382 + 5390 = 7772$$

$$b_{t_5} = 0.2 \times (7772 - 7500) + (0.8) \times 200 = 54.4 + 160 = 214.4$$

$$s_{t_5} = 0.4 \times (8240 - 7500 - 200) + (0.6) \times 300 = 215.2 + 180 = 395.2$$

$$\hat{y}_{t_6} = 7772 + 214.4 + 300 = 8286.4$$

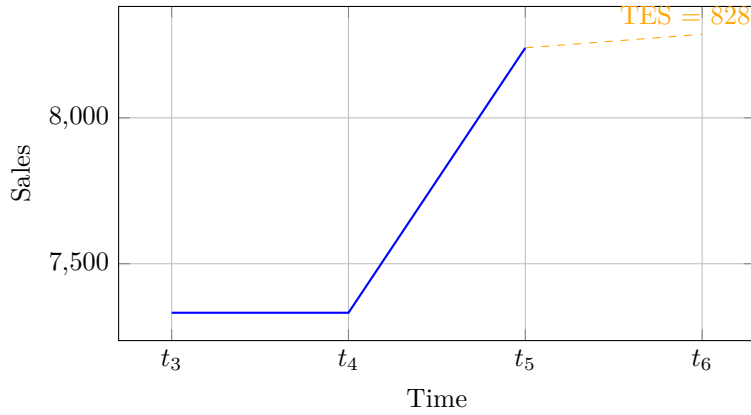


Figure 8: TES forecast with level, trend, and seasonality.

4.7 ARIMA Models

ARIMA (AutoRegressive Integrated Moving Average) is used for stationary series:

- **AR (AutoRegressive):**

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \epsilon_t$$

where ϕ_i are coefficients, ϵ_t is white noise.

- **I (Integrated):** Differencing order to achieve stationarity.
- **MA (Moving Average):**

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots$$

where θ_i are coefficients.

ACF and PACF:

- **ACF:** Correlation of y_t with y_{t-h} .
- **PACF:** Correlation after removing effects of earlier lags.

Example (AR(1)): For $y_t = 0.5y_{t-1} + \epsilon_t$, $\epsilon_t \sim N(0, 1)$:

$$y(t_5) = 8240, \quad \hat{y}_{t_6} = 0.5 \times 8240 = 4120$$

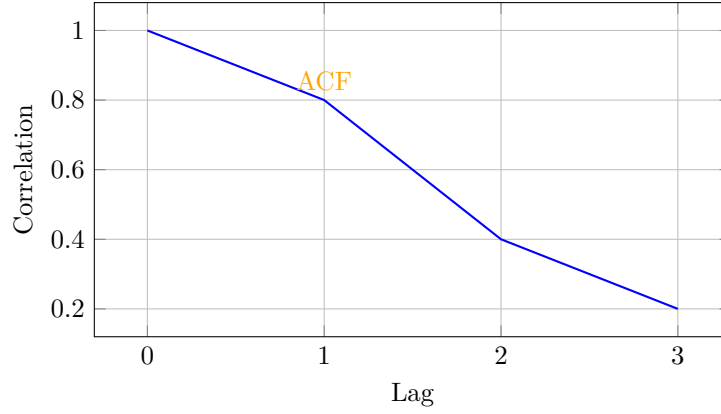


Figure 9: Auto-correlation function (ACF) for lag analysis.

5 Case Study: MobiPlus Sales Forecasting

5.1 Exploratory Data Analysis (EDA)

- Analyze 217 months of data (2001–2019).
- Set date as the index.
- Visualize to identify trends (increasing sales), seasonality (e.g., yearly cycles), and anomalies.

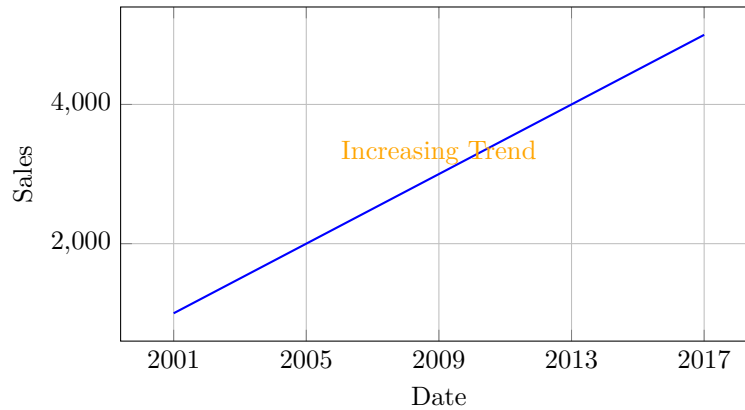


Figure 10: Sales data with trend and potential seasonality.

5.2 Handling Anomalies and Missing Values

- Use linear interpolation or centered MA for missing values.
- Retain outliers to preserve time series continuity, adjusting via imputation if necessary.

5.3 Forecasting Approach

- **Model Selection:** Use SES for stable series, DES for trending series, TES for seasonal series, or ARIMA for complex patterns.
- **Validation:** Ensure $\text{MAPE} \leq 5\%$ using:

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100$$

- **Range Forecast:** Generate confidence intervals using TES or ARIMA (e.g., $\pm 1.96 \times \text{std. error}$ for 95% CI).

6 Conclusion

MobiPlus can achieve accurate sales forecasts by:

- Handling missing values and anomalies with interpolation or MA.
- Decomposing the series to understand trends and seasonality.
- Applying SES, DES, TES, or ARIMA to meet $\text{MAPE} \leq 5\%$.
- Providing range forecasts for robust decision-making.